



```
In [2]: %pip install gdown
from collections import defaultdict
import json, math, gdown
import numpy as np
import pandas as pd
import plotly.express as px
from tqdm import tqdm
import requests
pd.options.display.float_format = '{:.2f}'.format
```

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: gdown in c:\users\mbaoj\appdata\roaming\python\python313\site-packages (6.0.0)
Requirement already satisfied: beautifulsoup4 in c:\programdata\anaconda3\lib\site-packages (from gdown) (4.13.5)
Requirement already satisfied: filelock in c:\programdata\anaconda3\lib\site-packages (from gdown) (3.20.0)
Requirement already satisfied: requests[socks] in c:\programdata\anaconda3\lib\site-packages (from gdown) (2.32.5)
Requirement already satisfied: tqdm in c:\programdata\anaconda3\lib\site-packages (from gdown) (4.67.1)
Requirement already satisfied: soupsieve>1.2 in c:\programdata\anaconda3\lib\site-packages (from beautifulsoup4->gdown) (2.5)
Requirement already satisfied: typing-extensions>=4.0.0 in c:\programdata\anaconda3\lib\site-packages (from beautifulsoup4->gdown) (4.15.0)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\programdata\anaconda3\lib\site-packages (from requests[socks]->gdown) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in c:\programdata\anaconda3\lib\site-packages (from requests[socks]->gdown) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\programdata\anaconda3\lib\site-packages (from requests[socks]->gdown) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in c:\programdata\anaconda3\lib\site-packages (from requests[socks]->gdown) (2026.2.25)
Requirement already satisfied: PySocks!=1.5.7,>=1.5.6 in c:\programdata\anaconda3\lib\site-packages (from requests[socks]->gdown) (1.7.1)
Requirement already satisfied: colorama in c:\programdata\anaconda3\lib\site-packages (from tqdm->gdown) (0.4.6)
Note: you may need to restart the kernel to use updated packages.

```
In [3]: # we use the latest data
url = "https://storage.googleapis.com/arena_external_data/public/clean_battle_response = requests.get(url)

with open('local_file_name.json', 'wb') as file:
    file.write(response.content)

# load the JSON data from the local file
with open('local_file_name.json', 'r') as file:
    battles = pd.read_json(file).sort_values(ascending=True, by=["tstamp"])
```

```
In [4]: # we use anony battles only for leaderboard
battles = battles[battles["anony"] == True]

# we de-duplicate top 0.1% redudant prompts
```

```
# see https://lmsys.org/blog/2024-05-17-category-hard/#note-enhancing-quality
print("Before dedup: ", len(battles))
battles = battles[battles["dedup_tag"].apply(lambda x: x.get("sampled", False))]
print("After dedup: ", len(battles))
```

Before dedup: 1799991

After dedup: 1670250

```
In [5]: fig = px.bar(battles["winner"].value_counts(),
                    title="Counts of Battle Outcomes", text_auto=True, height=400)
fig.update_layout(xaxis_title="Battle Outcome", yaxis_title="Count",
                  showlegend=False)
fig
```

```
In [6]: battles_no_ties = battles[~battles["winner"].str.contains("tie")]

fig = px.bar(pd.concat([battles["model_a"], battles["model_b"]]).value_counts(
              title="Battle Count for Each Model", text_auto=True)
fig.update_layout(xaxis_title="model", yaxis_title="Battle Count", height=400,
                  showlegend=False)
fig
```

```
In [7]: def visualize_battle_count(battles, title, show_num_models=30):
        ptbl = pd.pivot_table(battles, index="model_a", columns="model_b", aggfunc=
                               fill_value=0)
        battle_counts = ptbl + ptbl.T
        ordering = battle_counts.sum().sort_values(ascending=False).index
        ordering = ordering[:show_num_models]
        fig = px.imshow(battle_counts.loc[ordering, ordering],
                        title=title, text_auto=True)
        fig.update_layout(xaxis_title="Model B",
                          yaxis_title="Model A",
                          xaxis_side="top", height=800, width=800,
                          title_y=0.07, title_x=0.5,
                          font=dict(size=10))
        fig.update_traces(hovertemplate=
                          "Model A: {y}<br>Model B: {x}<br>Count: {z}<extra></e
        return fig

fig = visualize_battle_count(battles, title="Battle Count of Each Combination
fig
```

```
In [8]: visualize_battle_count(battles_no_ties, "Battle Count for Each Combination of
```

```
In [9]: visualize_battle_count(battles_no_ties, "Battle Count for Each Combination of
```

```
In [10]: lang_count = battles["language"].value_counts()
lang_count = lang_count.drop(index=("unknown"))
```

```
In [11]: topk = 15
fig = px.bar(lang_count.head(topk),
```

```

        title=f"Battle Counts for the Top {topk} Languages",
        text_auto=True, height=400, log_y=True)
fig.update_layout(xaxis_title="Language", yaxis_title="Count", showlegend=False)
fig

```

```

In [12]: fig = px.histogram(battles["turn"],
        title=f"Number of Conversation Turns",
        text_auto=True, height=400, log_y=True)
fig.update_layout(xaxis_title="Turns", yaxis_title="Count", showlegend=False)
fig

```

```

In [13]: def compute_pairwise_win_fraction(battles, max_num_models=30):
    # Times each model wins as Model A
    a_win_ptbl = pd.pivot_table(
        battles[battles['winner'] == "model_a"],
        index="model_a", columns="model_b", aggfunc="size", fill_value=0)

    # Table counting times each model wins as Model B
    b_win_ptbl = pd.pivot_table(
        battles[battles['winner'] == "model_b"],
        index="model_a", columns="model_b", aggfunc="size", fill_value=0)

    # Table counting number of A-B pairs
    num_battles_ptbl = pd.pivot_table(battles,
        index="model_a", columns="model_b", aggfunc="size", fill_value=0)

    # Computing the proportion of wins for each model as A and as B
    # against all other models
    row_beats_col_freq = (
        (a_win_ptbl + b_win_ptbl.T) /
        (num_battles_ptbl + num_battles_ptbl.T)
    )

    # Arrange ordering according to proportion of wins
    prop_wins = row_beats_col_freq.mean(axis=1).sort_values(ascending=False)
    prop_wins = prop_wins[:max_num_models]
    model_names = list(prop_wins.keys())
    row_beats_col = row_beats_col_freq.loc[model_names, model_names]
    return row_beats_col

def visualize_pairwise_win_fraction(battles, title, max_num_models=30):
    row_beats_col = compute_pairwise_win_fraction(battles, max_num_models)
    fig = px.imshow(row_beats_col, color_continuous_scale='RdBu',
        text_auto=".2f", title=title)
    fig.update_layout(xaxis_title="Model B: Loser",
        yaxis_title="Model A: Winner",
        xaxis_side="top", height=900, width=900,
        title_y=0.07, title_x=0.5)
    fig.update_traces(hovertemplate=
        "Model A: %{y}<br>Model B: %{x}<br>Fraction of A Wins: %{z}<br>
    return fig

```

```
In [14]: fig = visualize_pairwise_win_fraction(battles_no_ties,
        title = "Fraction of Model A Wins for All Non-tied A vs. B Battles")
fig
```

```
In [15]: row_beats_col_freq = compute_pairwise_win_fraction(battles_no_ties)
fig = px.bar(row_beats_col_freq.mean(axis=1).sort_values(ascending=False),
            title="Average Win Rate Against All Other Models (Assuming Uniform)",
            text_auto=".2f")
fig.update_layout(yaxis_title="Average Win Rate", xaxis_title="Model",
                  showlegend=False)
fig
```

```
In [16]: def compute_online_elo(battles, K=4, SCALE=400, BASE=10, INIT_RATING=1000):
        rating = defaultdict(lambda: INIT_RATING)

        for rd, model_a, model_b, winner in battles[['model_a', 'model_b', 'winner']]:
            ra = rating[model_a]
            rb = rating[model_b]
            ea = 1 / (1 + BASE ** ((rb - ra) / SCALE))
            eb = 1 / (1 + BASE ** ((ra - rb) / SCALE))
            if winner == "model_a":
                sa = 1
            elif winner == "model_b":
                sa = 0
            elif winner == "tie" or winner == "tie (bothbad)":
                sa = 0.5
            else:
                raise Exception(f"unexpected vote {winner}")
            rating[model_a] += K * (sa - ea)
            rating[model_b] += K * (1 - sa - eb)

        # calibrate llama-13b to 800
        delta = (800 - rating["llama-13b"])
        for model in battles["model_a"].unique():
            rating[model] += delta

        return rating
```

```
In [17]: def pretty_print_model_ratings(ratings):
        df = pd.DataFrame([
            [n, ratings[n]] for n in ratings.keys()
        ], columns=["Model", "Elo rating"]).sort_values("Elo rating", ascending=False)
        # df["Elo rating"] = (df["Elo rating"] + 0.5).astype(int)
        df.index = df.index + 1
        return df

online_elo_ratings = compute_online_elo(battles)
pretty_print_model_ratings(online_elo_ratings)
```

Out[17]:

	Model	Elo rating
1	gemini-1.5-pro-api-0409-preview	1106.87
2	gemini-1.5-pro-exp-0801	1074.94
3	chatgpt-4o-latest	1073.74
4	gpt-3.5-turbo-0314	1051.02
5	bard-jan-24-gemini-pro	1041.26
...	...	...
125	chatglm3-6b	814.55
126	oasst-pythia-12b	812.39
127	llama-13b	800.00
128	fastchat-t5-3b	794.37
129	dolly-v2-12b	781.44

129 rows × 2 columns

```
In [18]: def pretty_print_two_ratings(ratings_1, ratings_2, column_names):
df = pd.DataFrame([
    [n, ratings_1[n], ratings_2[n]] for n in ratings_1.keys()
], columns=["Model", column_names[0], column_names[1]]).sort_values(column_names[1])
df[column_names[0]] = (df[column_names[0]] + 0.5).astype(int)
df[column_names[1]] = (df[column_names[1]] + 0.5).astype(int)
df.index = df.index + 1
return df

elo_mle_ratings_reverse = compute_online_elo(battles.iloc[::-1])
pretty_print_two_ratings(online_elo_ratings,
                        elo_mle_ratings_reverse,
                        column_names=["Elo rating", "Elo rating with reverse"])
```

Out[18]:

	Model	Elo rating	Elo rating with reverse order
1	gemini-1.5-pro-api-0409-preview	1107	1112
2	gemini-1.5-pro-exp-0801	1075	1086
3	chatgpt-4o-latest	1074	1151
4	gpt-3.5-turbo-0314	1051	1162
5	bard-jan-24-gemini-pro	1041	1155
...	...	...	...
125	chatglm3-6b	815	893
126	oasst-pythia-12b	812	907
127	llama-13b	800	800
128	fastchat-t5-3b	794	863
129	dolly-v2-12b	781	833

129 rows × 3 columns

```
In [19]: def compute_mle_elo(
df, SCALE=400, BASE=10, INIT_RATING=1000, sample_weight=None
):
    from sklearn.linear_model import LogisticRegression
    ptbl_a_win = pd.pivot_table(
        df[df["winner"] == "model_a"],
        index="model_a",
        columns="model_b",
        aggfunc="size",
        fill_value=0,
    )
    # if no tie, create a zero matrix
    if sum(df["winner"].isin(["tie", "tie (bothbad)"])) == 0:
        ptbl_tie = pd.DataFrame(0, index=ptbl_a_win.index, columns=ptbl_a_win.
    else:
        ptbl_tie = pd.pivot_table(
            df[df["winner"].isin(["tie", "tie (bothbad)"])],
            index="model_a",
            columns="model_b",
            aggfunc="size",
            fill_value=0,
        )
        ptbl_tie = ptbl_tie + ptbl_tie.T

    ptbl_b_win = pd.pivot_table(
        df[df["winner"] == "model_b"],
        index="model_a",
        columns="model_b",
        aggfunc="size",
        fill_value=0,
```

```

)
ptbl_win = ptbl_a_win * 2 + ptbl_b_win.T * 2 + ptbl_tie

models = pd.Series(np.arange(len(ptbl_win.index)), index=ptbl_win.index)

p = len(models)
X = np.zeros([p * (p - 1) * 2, p])
Y = np.zeros(p * (p - 1) * 2)

cur_row = 0
sample_weights = []
for m_a in ptbl_win.index:
    for m_b in ptbl_win.columns:
        if m_a == m_b:
            continue
        # if nan skip
        if math.isnan(ptbl_win.loc[m_a, m_b]) or math.isnan(ptbl_win.loc[m_b, m_a]):
            continue
        X[cur_row, models[m_a]] = +math.log(BASE)
        X[cur_row, models[m_b]] = -math.log(BASE)
        Y[cur_row] = 1.0
        sample_weights.append(ptbl_win.loc[m_a, m_b])

        X[cur_row + 1, models[m_a]] = math.log(BASE)
        X[cur_row + 1, models[m_b]] = -math.log(BASE)
        Y[cur_row + 1] = 0.0
        sample_weights.append(ptbl_win.loc[m_b, m_a])
        cur_row += 2
X = X[:cur_row]
Y = Y[:cur_row]

lr = LogisticRegression(fit_intercept=False, penalty=None, tol=1e-6)
lr.fit(X, Y, sample_weight=sample_weights)
elo_scores = SCALE * lr.coef_[0] + INIT_RATING
if "mixtral-8x7b-instruct-v0.1" in models.index:
    elo_scores += 1114 - elo_scores[models["mixtral-8x7b-instruct-v0.1"]]
return pd.Series(elo_scores, index=models.index).sort_values(ascending=False)

```

```

In [20]: elo_mle_ratings = compute_mle_elo(battles)
preety_print_model_ratings(elo_mle_ratings)

```

Out[20]:

	Model	Elo rating
1	chatgpt-4o-latest	1315.16
2	gemini-1.5-pro-exp-0801	1298.43
3	gpt-4o-2024-05-13	1285.83
4	gpt-4o-mini-2024-07-18	1274.85
5	claude-3-5-sonnet-20240620	1270.98
...	...	...
125	chatglm-6b	878.99
126	fastchat-t5-3b	868.29
127	stablelm-tuned-alpha-7b	839.69
128	dolly-v2-12b	822.08
129	llama-13b	798.92

129 rows × 2 columns

```
In [21]: def get_bootstrap_result(battles, func_compute_elo, num_round):
rows = []
for i in tqdm(range(num_round), desc="bootstrap"):
    rows.append(func_compute_elo(battles.sample(frac=1.0, replace=True)))
df = pd.DataFrame(rows)
return df[df.median().sort_values(ascending=False).index]
```

```
In [22]: BOOTSTRAP_ROUNDS = 100

np.random.seed(42)
bootstrap_elo_lu = get_bootstrap_result(battles, compute_mle_elo, BOOTSTRAP_ROUNDS)

bootstrap: 100%|██████████| 100/100 [12:47<00:00, 7.68s/it]
```

```
In [23]: def visualize_bootstrap_scores(df, title):
bars = pd.DataFrame(dict(
    lower = df.quantile(.025),
    rating = df.quantile(.5),
    upper = df.quantile(.975))).reset_index(names="model").sort_values("ra
bars['error_y'] = bars['upper'] - bars["rating"]
bars['error_y_minus'] = bars['rating'] - bars["lower"]
bars['rating_rounded'] = np.round(bars['rating'], 2)
fig = px.scatter(bars, x="model", y="rating", error_y="error_y",
                 error_y_minus="error_y_minus", text="rating_rounded",
                 title=title)
fig.update_layout(xaxis_title="Model", yaxis_title="Rating",
                 height=600)
return fig

fig = visualize_bootstrap_scores(bootstrap_elo_lu, "Bootstrap of MLE Elo Ratin
```


fig

```
In [24]: np.random.seed(42)
bootstrap_online_elo = get_bootstrap_result(battles, compute_online_elo, BOTS)
```

bootstrap: 100%|██████████| 100/100 [16:46<00:00, 10.07s/it]

```
In [25]: pretty_print_two_ratings(bootstrap_elo.lu.quantile(.5),
                                bootstrap_online_elo.quantile(.5),
                                column_names=["Bootstrap Median of BT", "Bootstrap Me
```

Out[25]:

	Model	Bootstrap Median of BT	Bootstrap Median of Online Elo
1	chatgpt-4o-latest	1315	1315
2	gemini-1.5-pro-exp-0801	1298	1295
3	gpt-4o-2024-05-13	1286	1288
4	gpt-4o-mini-2024-07-18	1274	1278
5	claude-3-5-sonnet-20240620	1271	1272
...	...	...	...
125	chatglm-6b	879	881
126	fastchat-t5-3b	868	870
127	stablelm-tuned-alpha-7b	839	839
128	dolly-v2-12b	822	822
129	llama-13b	799	800

129 rows × 3 columns

```
In [26]: fig = visualize_bootstrap_scores(bootstrap_online_elo, "Bootstrap of Online Elo")
fig
```

```
In [28]: def predict_win_rate(elo_ratings, SCALE=400, BASE=10, INIT_RATING=1000):
    names = sorted(list(elo_ratings.keys()))
    wins = defaultdict(lambda: defaultdict(lambda: 0))
    for a in names:
        for b in names:
            ea = 1 / (1 + BASE ** ((elo_ratings[b] - elo_ratings[a]) / SCALE))
            wins[a][b] = ea
            wins[b][a] = 1 - ea

    data = {
        a: [wins[a][b] if a != b else np.nan for b in names]
        for a in names
    }
```

```

df = pd.DataFrame(data, index=names)
df.index.name = "model_a"
df.columns.name = "model_b"
return df.T

```

```

In [29]: win_rate = predict_win_rate(dict(bootstrap_elo_lu.quantile(0.5)))
ordered_models = win_rate.mean(axis=1).sort_values(ascending=False).index
ordered_models = ordered_models[:30]
fig = px.imshow(win_rate.loc[ordered_models, ordered_models],
                color_continuous_scale='RdBu', text_auto=".2f",
                title="Predicted Win Rate Using Elo Ratings for Model A in an
fig.update_layout(xaxis_title="Model B",
                  yaxis_title="Model A",
                  xaxis_side="top", height=900, width=900,
                  title_y=0.07, title_x=0.5)
fig.update_traces(hovertemplate=
                  "Model A: %{y}<br>Model B: %{x}<br>Win Rate: %{z}<extra></ex
fig

```

```

In [30]: def sample_battle_even(battles, n_per_battle):
groups = battles.groupby(["model_a", "model_b"], as_index=False)
resampled = (groups
              .apply(lambda grp: grp.sample(n_per_battle, replace=True))
              .reset_index(drop=True))
return resampled

```

```

In [31]: num_samples = 50
battles_even = sample_battle_even(battles, num_samples)
pd.pivot_table(battles_even, index="model_a", columns="model_b", aggfunc="size

```

C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

Out[31]:

model_b	RWKV-4-Raven-14B	alpaca-13b	athene-70b-0725	jan-24-gemini-pro
model_a				
RWKV-4-Raven-14B	0	50	0	0
alpaca-13b	50	0	0	0
athene-70b-0725	0	0	0	0
bard-jan-24-gemini-pro	0	0	0	0
chatglm-6b	50	50	0	0
...	...	...	...	...
yi-large	0	0	50	0
yi-large-preview	0	0	50	0
zephyr-7b-alpha	0	0	0	0
zephyr-7b-beta	0	0	0	0
zephyr-orpo-141b-A35b-v0.1	0	0	0	0

129 rows × 129 columns

```
In [32]: # Sampling Battles Evenly
def get_bootstrap_even_sample(battles, n_per_battle, func_compute_elo, num_rounds):
    rows = []
    for n in tqdm(range(num_rounds), desc="sampling battles evenly"):
        resampled = sample_battle_even(battles, n_per_battle)
        rows.append(func_compute_elo(resampled))
    df = pd.DataFrame(rows)
    return df[df.median().sort_values(ascending=False).index]
```

```
In [33]: print("number of samples per battle pair:", num_samples)
bootstrap_even_lu = get_bootstrap_even_sample(battles, num_samples, compute_ml)
number of samples per battle pair: 50
```

sampling battles evenly: 0%| | 0/100 [00:00<?, ?it/s]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 1%| | 1/100 [00:06<10:48, 6.55s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 2%|| | 2/100 [00:12<10:35, 6.48s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 3%|| | 3/100 [00:19<10:23, 6.42s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 4%|| | 4/100 [00:25<10:07, 6.32s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 5%|| | 5/100 [00:30<09:16, 5.85s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 6%|██████████ | 6/100 [00:36<09:17, 5.93s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 7%|██████████ | 7/100 [00:43<09:27, 6.10s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 8%|██████████ | 8/100 [00:49<09:31, 6.22s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 9%|██████████ | 9/100 [00:56<09:33, 6.31s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 10%|██████████ | 10/100 [01:01<09:04, 6.05s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 11%|██████████ | 11/100 [01:08<09:11, 6.20s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 12%|██████████| 12/100 [01:14<09:06, 6.21s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 13%|██████████| 13/100 [01:20<09:10, 6.33s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 14%|██████████| 14/100 [01:26<08:33, 5.97s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 15%|██████████| 15/100 [01:32<08:30, 6.01s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 16%|██████████| 16/100 [01:38<08:30, 6.08s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 17%|██████████| 17/100 [01:44<08:31, 6.16s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 18%|██████████ | 18/100 [01:50<08:06, 5.93s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 19%|██████████ | 19/100 [01:56<08:09, 6.04s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 20%|██████████ | 20/100 [02:02<08:11, 6.14s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 21%|██████████ | 21/100 [02:09<08:14, 6.26s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 22%|██████████ | 22/100 [02:15<08:09, 6.27s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 23%|██████████ | 23/100 [02:20<07:37, 5.94s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 24%|██████████ | 24/100 [02:27<07:45, 6.12s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 25%|██████████ | 25/100 [02:33<07:50, 6.28s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 26%|██████████ | 26/100 [02:40<07:50, 6.36s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 27%|██████████ | 27/100 [02:45<07:20, 6.03s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 28%|██████████ | 28/100 [02:52<07:34, 6.32s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 29%|██████████ | 29/100 [02:59<07:43, 6.53s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 30%|██████████ | 30/100 [03:07<07:52, 6.76s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 31%|██████████ | 31/100 [03:13<07:30, 6.52s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 32%|██████████ | 32/100 [03:19<07:16, 6.43s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 33%|██████████ | 33/100 [03:25<07:09, 6.41s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 34%|██████████ | 34/100 [03:31<06:56, 6.31s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 35%|██████████ | 35/100 [03:36<06:22, 5.89s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 36%|██████ | 36/100 [03:42<06:25, 6.02s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 37%|██████ | 37/100 [03:49<06:27, 6.16s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 38%|██████ | 38/100 [03:55<06:25, 6.22s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 39%|██████ | 39/100 [04:02<06:34, 6.46s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 40%|██████ | 40/100 [04:07<06:04, 6.08s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 41%|██████ | 41/100 [04:13<05:55, 6.02s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 42%|██████ | 42/100 [04:20<05:53, 6.10s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 43%|██████ | 43/100 [04:26<05:50, 6.16s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 44%|██████ | 44/100 [04:32<05:48, 6.22s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 45%|██████ | 45/100 [04:38<05:41, 6.20s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 46%|██████ | 46/100 [04:44<05:20, 5.94s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 47%|██████ | 47/100 [04:50<05:20, 6.04s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 48%|██████████ | 48/100 [04:56<05:15, 6.06s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 49%|██████████ | 49/100 [05:02<05:08, 6.05s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 50%|██████████ | 50/100 [05:07<04:49, 5.79s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 51%|██████████ | 51/100 [05:14<04:53, 5.98s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 52%|██████████ | 52/100 [05:20<04:54, 6.13s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 53%|██████████ | 53/100 [05:27<04:49, 6.15s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 54%|██████ | 54/100 [05:33<04:45, 6.21s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 55%|██████ | 55/100 [05:38<04:26, 5.93s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 56%|██████ | 56/100 [05:44<04:23, 5.99s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 57%|██████ | 57/100 [05:51<04:22, 6.10s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 58%|██████ | 58/100 [05:57<04:20, 6.20s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 59%|██████ | 59/100 [06:03<04:16, 6.26s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 60%|██████ | 60/100 [06:08<03:54, 5.86s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 61%|██████ | 61/100 [06:14<03:50, 5.90s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 62%|██████ | 62/100 [06:21<03:49, 6.05s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 63%|██████ | 63/100 [06:27<03:42, 6.01s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 64%|██████ | 64/100 [06:32<03:31, 5.87s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 65%|██████ | 65/100 [06:38<03:27, 5.94s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 66%|██████ | 66/100 [06:45<03:27, 6.11s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 67%|██████ | 67/100 [06:51<03:24, 6.20s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 68%|██████ | 68/100 [06:57<03:10, 5.94s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 69%|██████ | 69/100 [07:03<03:08, 6.09s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 70%|██████ | 70/100 [07:09<03:04, 6.14s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 71%|██████ | 71/100 [07:18<03:22, 6.99s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 72%|██████████ | 72/100 [07:28<03:40, 7.88s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 73%|██████████ | 73/100 [07:40<04:01, 8.96s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 74%|██████████ | 74/100 [07:50<04:06, 9.49s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 75%|██████████ | 75/100 [08:01<04:06, 9.85s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 76%|██████████ | 76/100 [08:12<04:03, 10.13s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 77%|██████████ | 77/100 [08:23<03:58, 10.36s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 78%|██████████ | 78/100 [08:33<03:44, 10.22s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 79%|██████████ | 79/100 [08:45<03:46, 10.79s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 80%|██████████ | 80/100 [08:56<03:38, 10.91s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 81%|██████████ | 81/100 [09:07<03:30, 11.06s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 82%|██████████ | 82/100 [09:18<03:15, 10.87s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 83%|██████████ | 83/100 [09:29<03:08, 11.08s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 84%|██████████ | 84/100 [09:42<03:04, 11.50s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 85%|██████████ | 85/100 [09:54<02:53, 11.55s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 86%|██████████ | 86/100 [10:04<02:35, 11.09s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 87%|██████████ | 87/100 [10:18<02:36, 12.01s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 88%|██████████ | 88/100 [10:29<02:22, 11.85s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 89%|██████████ | 89/100 [10:41<02:08, 11.71s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 90%|██████████ | 90/100 [10:50<01:50, 11.07s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 91%|██████████ | 91/100 [11:01<01:40, 11.11s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 92%|██████████ | 92/100 [11:14<01:32, 11.52s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 93%|██████████ | 93/100 [11:29<01:27, 12.53s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 94%|██████████ | 94/100 [11:35<01:04, 10.77s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

sampling battles evenly: 95%|██████████ | 95/100 [11:42<00:48, 9.64s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

```
sampling battles evenly: 96%|██████████| 96/100 [11:49<00:34, 8.71s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:
```

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

```
sampling battles evenly: 97%|██████████| 97/100 [11:55<00:24, 8.07s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:
```

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

```
sampling battles evenly: 98%|██████████| 98/100 [12:01<00:14, 7.28s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:
```

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

```
sampling battles evenly: 99%|██████████| 99/100 [12:07<00:07, 7.03s/it]C:\Users\mbaoj\AppData\Local\Temp\ipykernel_25652\3725215937.py:4: FutureWarning:
```

DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

```
sampling battles evenly: 100%|██████████| 100/100 [12:14<00:00, 7.34s/it]
```

```
In [34]: fig = visualize_bootstrap_scores(bootstrap_even_lu, f"Bootstrap of MLE Elo Est  
fig
```

```
In [35]: english_only_battles = battles[battles["language"] == "English"]  
elo_ratings = compute_mle_elo(english_only_battles)  
pd.DataFrame(elo_ratings)
```

Out[35]:

0

model_a	
chatgpt-4o-latest	1293.96
gemini-1.5-pro-exp-0801	1266.56
gpt-4o-2024-05-13	1263.66
llama-3.1-405b-instruct	1256.88
gpt-4o-mini-2024-07-18	1255.61
...	...
fastchat-t5-3b	871.25
chatglm-6b	858.02
stablelm-tuned-alpha-7b	830.05
dolly-v2-12b	804.18
llama-13b	778.58

129 rows × 1 columns

```
In [36]: non_english_battles = battles[battles["language"] != "English"]
elo_ratings = compute_mle_elo(non_english_battles)
pd.DataFrame(elo_ratings)
```

Out[36]:

0

model_a	
chatgpt-4o-latest	1349.54
gemini-1.5-pro-exp-0801	1344.25
gpt-4o-2024-05-13	1319.00
gemini-advanced-0514	1309.92
claude-3-5-sonnet-20240620	1308.96
...	...
oasst-pythia-12b	880.54
dolly-v2-12b	860.85
llama-13b	848.71
stablelm-tuned-alpha-7b	843.59
fastchat-t5-3b	811.91

129 rows × 1 columns